UNIT 8 · ARS/NET PATTERN

MCQ Bank — 105 Questions with Answers

Ferrous, non-ferrous, alloy and composite materials (Q1–Q35)

Q1. Ferrous materials are based on:

(a) Aluminium (b) Iron
(c) Copper (d) Zinc

Q2. As carbon content of plain-carbon steel increases:

(a) Hardness rises, ductility falls (b) Both rise
(c) Both fall (d) No change

Q3. Mild (low-carbon) steel has carbon content:

(a) <0.3% (b) 0.3–0.6%
(c) 0.6–1.0% (d) 2–4%

Q4. High-carbon steel (0.6–1.4% C) is typically used for:

(a) Body panels (b) Plough shares, springs, cutting edges
(c) Radiators (d) Wiring

Q5. Cast iron contains carbon in the range:

(a) 0.1–0.3% (b) 0.3–0.6%
(c) 2–4% (d) 6–8%

Q6. Grey cast iron is valued for:

(a) High ductility (b) Machinability, damping, compressive strength
(c) Weldability (d) Corrosion resistance

Q7. The hardest and most brittle cast iron is:

(a) Grey (b) White
(c) Malleable (d) SG/ductile

Q8. Spheroidal-graphite (ductile) cast iron is made by treating the melt with:

(a) Sulphur (b) Magnesium
(c) Lead (d) Zinc

Q9. Which alloying element most increases HARDENABILITY at tiny additions?

(a) Boron (b) Lead
(c) Tin (d) Zinc

Q10. Chromium above about 11% makes steel:

(a) Cast iron (b) Stainless (corrosion-resistant)
(c) Brittle only (d) Non-magnetic wire

Q11. Manganese (Hadfield) steel is chosen for:

(a) Electrical use (b) High-impact abrasive wear (work-hardening)
(c) Light weight (d) Radiators

Q12. Nickel is added to steel mainly to improve:

(a) Toughness (b) Density
(c) Colour (d) Conductivity

Q13. Pig iron is produced in a:

(a) Cupola
(b) Blast furnace
(c) Arc furnace
(d) Ladle

Q14. Steel is made from pig iron/scrap in a:

(a) Blast furnace
(b) Basic Oxygen or Electric Arc Furnace
(c) Cupola
(d) Kiln

Q15. Cast iron for foundry work is commonly melted in a:

(a) Blast furnace
(b) Cupola
(c) Reverberatory only
(d) Rotary kiln

Q16. Aluminium's density is about:

(a) 2.7 g/cm³ (⅓ of steel)
(b) 7.8 g/cm³
(c) 11 g/cm³
(d) 19 g/cm³

Q17. Aluminium alloys are used in tractors for:

(a) Plough shares
(b) Radiators, pistons, cylinder heads
(c) Crankshafts
(d) Springs

Q18. Brass is an alloy of:

(a) Cu + Sn
(b) Cu + Zn
(c) Fe + C
(d) Al + Mg

Q19. Bronze is an alloy of:

(a) Cu + Zn
(b) Cu + Sn
(c) Fe + Ni
(d) Al + Si

Q20. Bronze is preferred for bushes and bearings because of its:

(a) Magnetism
(b) Good wear resistance and low friction
(c) Light weight
(d) Transparency

Q21. Galvanising for corrosion protection uses a coating of:

(a) Zinc
(b) Lead
(c) Tin
(d) Nickel

Q22. Copper is chosen for windings and radiator tubes because it is a good:

(a) Insulator
(b) Electrical/thermal conductor
(c) Abrasive
(d) Magnet

Q23. A composite material consists of:

(a) One pure metal
(b) A matrix plus a reinforcement
(c) Two gases
(d) Only polymer

Q24. FRP (fibre-reinforced polymer) is used in farm machinery for:

(a) Crankshaft
(b) Sprayer/seed tanks, guards, hoods
(c) Gears
(d) Bearings

Q25. The chief advantage of composites is:

(a) Highest density
(b) High strength-to-weight and corrosion resistance
(c) Low cost always
(d) Easy recycling

Q26. Tungsten-carbide particles in a metal matrix (hard-facing/cermet) are used for:

(a) Light panels
(b) Extreme wear resistance on tool edges
(c) Wiring
(d) Cushioning

Q27. The main duty a plough share material must resist is:

(a) Electrical load
(b) Abrasive soil wear + impact
(c) Heat only
(d) Light

Q28. Target hardness for a hardened plough share/point is about:

(a) 20–25 HRC
(b) 35–40 HRC
(c) 50–60 HRC
(d) 90 HRC

Q29. Boron steel grade 10B36 contains roughly:

(a) 0.32–0.37% C with ~0.0005%+ boron
(b) 2% C
(c) 12% Cr
(d) 30% Ni

Q30. Abrasion-resistant steel AR400 has a Brinell hardness of about:

(a) 100
(b) 200
(c) 360–440
(d) 700

Q31. The hardness test using a diamond pyramid indenter is:

(a) Brinell
(b) Rockwell
(c) Vickers
(d) Izod

Q32. The Rockwell C scale (HRC) uses a:

(a) Steel ball only
(b) Diamond cone indenter
(c) Flat punch
(d) Pendulum

Q33. Impact toughness of a material is measured by the:

(a) Tensile test
(b) Charpy/Izod impact test
(c) Hardness test
(d) Fatigue test

Q34. The process that actually develops the required hardness/toughness in a chosen steel is:

(a) Painting
(b) Heat treatment (hardening + tempering, case-hardening)
(c) Galvanising
(d) Machining

Q35. Case-hardening (carburising) gives a component:

(a) Soft surface, hard core
(b) Hard wear-resistant surface with a tough core
(c) Uniform softness
(d) Corrosion coating

Answers

Q1 — (b) Iron
Q2 — (a) Hardness rises, ductility falls — *Carbon raises hardness/strength but lowers ductility and weldability.*
Q3 — (a) <0.3%
Q4 — (b) Plough shares, springs, cutting edges — *Hence high-C steel for shares/springs, mild steel for panels.*
Q5 — (c) 2–4%
Q6 — (b) Machinability, damping, compressive strength — *Grey CI — machinable, damping.*
Q7 — (b) White — *White CI — hard and brittle (cementite).*
Q8 — (b) Magnesium — *SG/ductile CI — Mg-treated nodules → tough.*
Q9 — (a) Boron — *Boron gives huge hardenability at ~0.0005%+.*
Q10 — (b) Stainless (corrosion-resistant)
Q11 — (b) High-impact abrasive wear (work-hardening)
Q12 — (a) Toughness
Q13 — (b) Blast furnace
Q14 — (b) Basic Oxygen or Electric Arc Furnace
Q15 — (b) Cupola
Q16 — (a) 2.7 g/cm³ (⅓ of steel)

Q17 — (b) Radiators, pistons, cylinder heads
Q18 — (b) Cu + Zn — *Brass = Cu + Zn (fittings).*
Q19 — (b) Cu + Sn — *Bronze = Cu + Sn.*
Q20 — (b) Good wear resistance and low friction — *Bronze for bearings/bushes — low friction.*
Q21 — (a) Zinc
Q22 — (b) Electrical/thermal conductor
Q23 — (b) A matrix plus a reinforcement
Q24 — (b) Sprayer/seed tanks, guards, hoods
Q25 — (b) High strength-to-weight and corrosion resistance
Q26 — (b) Extreme wear resistance on tool edges
Q27 — (b) Abrasive soil wear + impact
Q28 — (c) 50–60 HRC — *Shares are hardened to 50–60 HRC by heat treatment.*
Q29 — (a) 0.32–0.37% C with ~0.0005%+ boron — *10B36 ≈ 0.32–0.37% C with boron — as given in the source notes.*
Q30 — (c) 360–440 — *AR grades scale by BHN — AR400 ≈ 360–440.*
Q31 — (c) Vickers
Q32 — (b) Diamond cone indenter
Q33 — (b) Charpy/Izod impact test
Q34 — (b) Heat treatment (hardening + tempering, case-hardening) — *Heat treatment develops the hardness/toughness of a chosen steel.*
Q35 — (b) Hard wear-resistant surface with a tough core — *Case-hardening: hard surface + tough core — ideal for wear parts.*

Workshop practice, press operations, machining, jigs, fixtures and gauges (Q36–Q70)

Q36. The correct order of prototype fabrication is:

(a) Machining → design → cutting
(b) Design → marking → cutting → forming → joining → machining → assembly
(c) Assembly first
(d) Welding → design

Q37. A tool used to cut internal threads is a:

(a) Die
(b) Tap
(c) Reamer
(d) Scriber

Q38. External threads on a rod are cut with a:

(a) Tap
(b) Die
(c) Reamer
(d) Broach

Q39. A reamer is used to:

(a) Drill a new hole
(b) Finish/size an existing hole accurately
(c) Cut threads
(d) Mark layout

Q40. In BLANKING, the useful product is the:

(a) Sheet with the hole
(b) Piece punched out
(c) Scrap slug
(d) Chip

Q41. In PIERCING/PUNCHING, the removed slug is:

(a) The product
(b) Scrap
(c) Reused as tool
(d) The die

Q42. Shearing produces a:

(a) Cup
(b) Straight-line cut separating the sheet
(c) Threaded hole
(d) Weld

Q43. Making a cup or shell from a flat blank is:

(a) Blanking
(b) Deep drawing
(c) Piercing
(d) Coining

Q44. Cutting force in blanking is F =

(a) $L \times t \times \tau_s$
(b) L/t
(c) t/τ_s
(d) $L + t + \tau_s$

Q45. Punch–die clearance in cutting is typically:

(a) 0%
(b) 5–10% of thickness per side
(c) 50%
(d) Equal to thickness

Q46. A compound die performs:

(a) One operation
(b) Two operations (e.g., blank + pierce) in one stroke
(c) Only bending
(d) No cutting

Q47. Machining removes material in the form of a:

(a) Weld bead
(b) Chip
(c) Casting
(d) Powder only

Q48. Increasing the rake angle of a cutting tool generally:

(a) Increases cutting force
(b) Decreases cutting force but weakens the edge
(c) No effect
(d) Stops chip flow

Q49. The clearance (relief) angle prevents:

(a) Chip curling
(b) The tool flank rubbing the workpiece
(c) Overheating only
(d) Vibration

Q50. A built-up edge (BUE) tends to form when machining:

(a) Brittle metals
(b) Ductile metals at certain speeds
(c) Ceramics
(d) Glass

Q51. Cylindrical parts are produced mainly by:

(a) Milling
(b) Turning (lathe)
(c) Grinding only
(d) Shaping

Q52. Enlarging and finishing an existing hole is:

(a) Drilling
(b) Boring
(c) Blanking
(d) Knurling

Q53. Flat surfaces and slots are commonly produced by:

(a) Turning
(b) Milling
(c) Reaming
(d) Drawing

Q54. The three basic cutting parameters are:

(a) Speed, feed, depth of cut
(b) Voltage, current, time
(c) Load, area, stress
(d) Mass, velocity, force

Q55. In ascending order of cutting performance the tool materials are:

(a) Carbide < HSS < steel
(b) HSS < carbide < ceramic < CBN/diamond
(c) Diamond < HSS
(d) Ceramic < steel

Q56. Taylor's tool life equation is:

(a) $V \cdot T^n = C$ (b) $V + T = C$

(c) $V/T = C$ (d) $V \cdot T = n$

Q57. In $V \cdot T^n = C$, V and T are:

(a) Volume and torque (b) Cutting speed and tool life

(c) Voltage and time (d) Velocity and temperature

Q58. The exponent n for HSS tools is about:

(a) 0.16 (b) 0.30

(c) 0.40 (d) 0.60

Q59. The exponent n for ceramic tools is about:

(a) 0.10 (b) 0.16

(c) 0.30 (d) 0.40

Q60. A higher value of n means tool life is:

(a) More sensitive to speed (b) Less sensitive to cutting speed

(c) Zero (d) Infinite

Q61. With n = 0.2, raising speed from 40 to 60 m/min changes tool life from 60 min to about:

(a) 40 min (b) 20 min

(c) 8 min (d) 60 min

Q62. Common tool wear modes are:

(a) Flank and crater wear (b) Rust only

(c) Melting (d) Corrosion only

Q63. A JIG differs from a fixture in that it also:

(a) Holds the tool only (b) Guides the cutting tool (via bushes)

(c) Measures parts (d) Welds

Q64. A FIXTURE is typically used for:

(a) Drilling with guide bushes (b) Milling, turning, welding (holds work, no tool guidance)

(c) Threading (d) Marking

Q65. Jigs are most associated with which operation?

(a) Milling (b) Turning

(c) Drilling/reaming/tapping (d) Grinding

Q66. The location principle used to constrain a workpiece fully is:

(a) 3-2-1 (six-point) (b) 1-1-1

(c) 5-5-5 (d) 2-2-2

Q67. A GO / NO-GO gauge is a type of:

(a) Indicating gauge (b) Limit gauge

(c) Slip gauge (d) Radius gauge

Q68. A plug gauge is used to check:

(a) Shaft diameters (b) Hole (internal) diameters

(c) Thread pitch (d) Radius

Q69. Snap and ring gauges check:

(a) Holes (b) External (shaft) dimensions

(c) Angles (d) Flatness

Q70. By Taylor's principle of gauging, the GO gauge checks:

(a) Minimum-metal, one dimension
(b) Maximum-metal condition on all dimensions together
(c) Only length
(d) Surface finish

Answers

Q36 — (b) Design → marking → cutting → forming → joining → machining → assembly
Q37 — (b) Tap
Q38 — (b) Die
Q39 — (b) Finish/size an existing hole accurately
Q40 — (b) Piece punched out — *Blanking → the punched-out piece is the product.*
Q41 — (b) Scrap — *Piercing → the hole is wanted; the slug is scrap — the classic pair.*
Q42 — (b) Straight-line cut separating the sheet
Q43 — (b) Deep drawing
Q44 — (a) L × t × τ_s — *$F = L \cdot t \cdot \tau_s$.*
Q45 — (b) 5–10% of thickness per side — *Clearance 5–10% of thickness per side controls edge quality.*
Q46 — (b) Two operations (e.g., blank + pierce) in one stroke
Q47 — (b) Chip
Q48 — (b) Decreases cutting force but weakens the edge
Q49 — (b) The tool flank rubbing the workpiece
Q50 — (b) Ductile metals at certain speeds
Q51 — (b) Turning (lathe)
Q52 — (b) Boring
Q53 — (b) Milling
Q54 — (a) Speed, feed, depth of cut
Q55 — (b) HSS < carbide < ceramic < CBN/diamond
Q56 — (a) $V \cdot T^n = C$ — *$V \cdot T^n = C$.*
Q57 — (b) Cutting speed and tool life
Q58 — (a) 0.16 — *n(HSS) ≈ 0.16 in the source (textbook ranges vary; the order is what is tested).*
Q59 — (d) 0.40 — *n(ceramic) ≈ 0.40 in the source.*
Q60 — (b) Less sensitive to cutting speed — *Higher n = life less speed-sensitive.*
Q61 — (c) 8 min — *$T_2 = 60 \cdot (40/60)^{(1/0.2)} = 60 \cdot (0.667)^5 \approx 7.9$ min.*
Q62 — (a) Flank and crater wear
Q63 — (b) Guides the cutting tool (via bushes) — *A JIG guides the tool (drill bushes).*
Q64 — (b) Milling, turning, welding (holds work, no tool guidance) — *A FIXTURE only holds (milling/turning).*
Q65 — (c) Drilling/reaming/tapping — *Jigs: drilling, reaming, tapping.*
Q66 — (a) 3-2-1 (six-point) — *3-2-1 six-point location.*
Q67 — (b) Limit gauge — *GO/NO-GO = limit gauge.*
Q68 — (b) Hole (internal) diameters — *Plug gauge — holes.*
Q69 — (b) External (shaft) dimensions — *Snap/ring gauges — shafts.*
Q70 — (b) Maximum-metal condition on all dimensions together — *Taylor's principle — GO checks max-metal on all dimensions; NO-GO checks min-metal one dimension at a time.*

Casting, welding, weldment testing and workshop planning (Q71–Q105)

Q71. In sand casting, the top half of the moulding flask is the:

(a) Drag
(b) Cope
(c) Core
(d) Riser

Q72. The bottom half of the flask is the:

(a) Cope
(b) Drag
(c) Sprue
(d) Gate

Q73. Molten metal is poured into the mould through the:

(a) Riser
(b) Sprue/gating system
(c) Core
(d) Vent

Q74. A riser in a mould serves to:

(a) Vent gases only
(b) Feed metal to compensate shrinkage
(c) Locate the core
(d) Cool the metal

Q75. Internal cavities in a casting are produced using a:

(a) Riser
(b) Core
(c) Chill
(d) Sprue

Q76. Which is NOT a pattern allowance?

(a) Shrinkage
(b) Draft
(c) Machining
(d) Porosity

Q77. Draft allowance is provided to:

(a) Add strength
(b) Allow easy withdrawal of the pattern
(c) Increase weight
(d) Cool faster

Q78. Die casting is generally limited to:

(a) Ferrous metals
(b) Non-ferrous metals (Al, Zn, Mg)
(c) Ceramics
(d) Wood

Q79. Compared with sand casting, die casting gives:

(a) Lower accuracy
(b) Higher accuracy and finish, high production rate
(c) Cheaper tooling
(d) Larger castings only

Q80. Sand casting is preferred over die casting for:

(a) Mass production of small Al parts
(b) Large parts, ferrous metals, prototypes (low tooling cost)
(c) Zinc toys
(d) Thin wall precision

Q81. Lost-wax precision casting is also called:

(a) Shell moulding
(b) Investment casting
(c) Centrifugal casting
(d) Die casting

Q82. A casting defect from trapped gas is:

(a) Hot tear
(b) Blowhole/porosity
(c) Misrun
(d) Mismatch

Q83. Incomplete filling of the mould cavity gives:

(a) Shrinkage cavity
(b) Misrun/cold shut
(c) Inclusion
(d) Warping

Q84. Welding that joins metals by melting the joint is:

(a) Pressure welding
(b) Fusion welding
(c) Soldering
(d) Riveting

Q85. Oxy-acetylene gas welding flame temperature is about:

(a) 800 °C
(b) 1500 °C
(c) 3200 °C
(d) 6000 °C

Q86. SMAW commonly refers to:

(a) TIG
(b) Stick/manual metal arc welding with coated electrode
(c) MIG
(d) Submerged arc

Q87. In TIG (GTAW) welding the electrode is:

(a) Consumable wire
(b) Non-consumable tungsten
(c) Carbon rod
(d) Flux tube

Q88. The shielding gas/flux in arc welding protects the pool from:

(a) Light
(b) Atmospheric oxygen and nitrogen
(c) Water
(d) Dust

Q89. Brazing uses a filler that melts:

(a) Below 450 °C
(b) Above 450 °C but below base-metal melting
(c) Same as base metal
(d) Never

Q90. Soldering filler melts at:

(a) Below 450 °C
(b) Above 1000 °C
(c) 3200 °C
(d) Same as steel

Q91. Resistance spot welding produces heat by:

(a) A flame
(b) Electric current + pressure at the contact
(c) Friction
(d) Chemical reaction

Q92. The base metal region altered by welding heat but not melted is the:

(a) Weld metal
(b) Heat-affected zone (HAZ)
(c) Slag
(d) Flux

Q93. The HAZ in conventional arc welds typically extends about:

(a) 0.1 mm
(b) 5–15 mm
(c) 50 mm
(d) 200 mm

Q94. The most crack-prone region of a weldment is often the:

(a) Base metal far away
(b) HAZ
(c) Filler spool
(d) Flux

Q95. Cracks that occur later in service under load are called:

(a) Hot cracks
(b) Cold (hydrogen) cracks
(c) Solidification cracks
(d) Craters

Q96. Undercut is a defect that is a:

(a) Gas pore
(b) Groove melted into the base metal at the weld toe
(c) Slag pocket
(d) Crack

Q97. Distortion in weldments is mainly due to:

(a) Residual (thermal) stresses
(b) Painting
(c) Machining
(d) Galvanising

Q98. Which is a NON-destructive test?

(a) Tensile test
(b) Radiography
(c) Bend test
(d) Charpy impact

Q99. Dye-penetrant testing (PT) detects:

(a) Deep internal flaws
(b) Surface-breaking defects on any metal
(c) Composition
(d) Hardness

Q100. Magnetic-particle testing (MT) is limited to:

(a) Non-magnetic metals
(b) Ferromagnetic metals (surface/near-surface)
(c) Plastics
(d) Gases

Q101. Internal defects with a permanent film record are best found by:

(a) Visual
(b) Radiography (X/γ-ray)
(c) Penetrant
(d) Hardness

Q102. Ultrasonic testing (UT) detects flaws using:

(a) X-rays
(b) Reflected high-frequency sound
(c) Dye
(d) Magnetism

Q103. A basic machine for producing cylindrical parts in a workshop is the:

(a) Lathe
(b) Shear
(c) Press-brake
(d) Welding set

Q104. The primary aim of good workshop layout is:

(a) Maximum back-tracking
(b) Smooth material flow with minimum handling and idle time
(c) More noise
(d) Higher scrap

Q105. In workshop planning, forging and welding areas should be:

(a) Next to precision machining and inspection
(b) Separated from precision machining/inspection (heat, noise, fumes)
(c) In the office
(d) Removed entirely

Answers

Q71 — (b) Cope — *Sand-casting anatomy — cope is the top half.*
Q72 — (b) Drag — *Drag is the bottom half.*
Q73 — (b) Sprue/gating system
Q74 — (b) Feed metal to compensate shrinkage — *Riser feeds shrinkage.*
Q75 — (b) Core — *Core forms internal cavities.*
Q76 — (d) Porosity — *Porosity is a DEFECT, not an allowance.*
Q77 — (b) Allow easy withdrawal of the pattern — *Draft allows easy pattern withdrawal.*
Q78 — (b) Non-ferrous metals (Al, Zn, Mg) — *Die casting = permanent steel die, high pressure, non-ferrous.*
Q79 — (b) Higher accuracy and finish, high production rate — *Mass production with high accuracy.*
Q80 — (b) Large parts, ferrous metals, prototypes (low tooling cost) — *Sand casting = expendable mould, any metal, large parts, cheap tooling.*
Q81 — (b) Investment casting
Q82 — (b) Blowhole/porosity
Q83 — (b) Misrun/cold shut
Q84 — (b) Fusion welding
Q85 — (c) 3200 °C
Q86 — (b) Stick/manual metal arc welding with coated electrode
Q87 — (b) Non-consumable tungsten
Q88 — (b) Atmospheric oxygen and nitrogen
Q89 — (b) Above 450 °C but below base-metal melting — *Brazing filler > 450 °C.*
Q90 — (a) Below 450 °C — *Soldering filler < 450 °C — the standard temperature line.*
Q91 — (b) Electric current + pressure at the contact
Q92 — (b) Heat-affected zone (HAZ) — *HAZ is unmelted but altered.*
Q93 — (b) 5–15 mm — *HAZ ~5–15 mm in conventional arc welds.*
Q94 — (b) HAZ — *Often the weakest, most crack-prone region.*

Q95 — (b) Cold (hydrogen) cracks — *Hot cracks form on cooling; cold/hydrogen cracks appear later in service.*
Q96 — (b) Groove melted into the base metal at the weld toe
Q97 — (a) Residual (thermal) stresses
Q98 — (b) Radiography — *Radiography is non-destructive.*
Q99 — (b) Surface-breaking defects on any metal — *PT — surface flaws, any metal.*
Q100 — (b) Ferromagnetic metals (surface/near-surface) — *MT — ferrous surface/near-surface only.*
Q101 — (b) Radiography (X/γ-ray) — *RT — internal, permanent film.*
Q102 — (b) Reflected high-frequency sound — *UT — internal, reflected sound.*
Q103 — (a) Lathe
Q104 — (b) Smooth material flow with minimum handling and idle time
Q105 — (b) Separated from precision machining/inspection (heat, noise, fumes)
